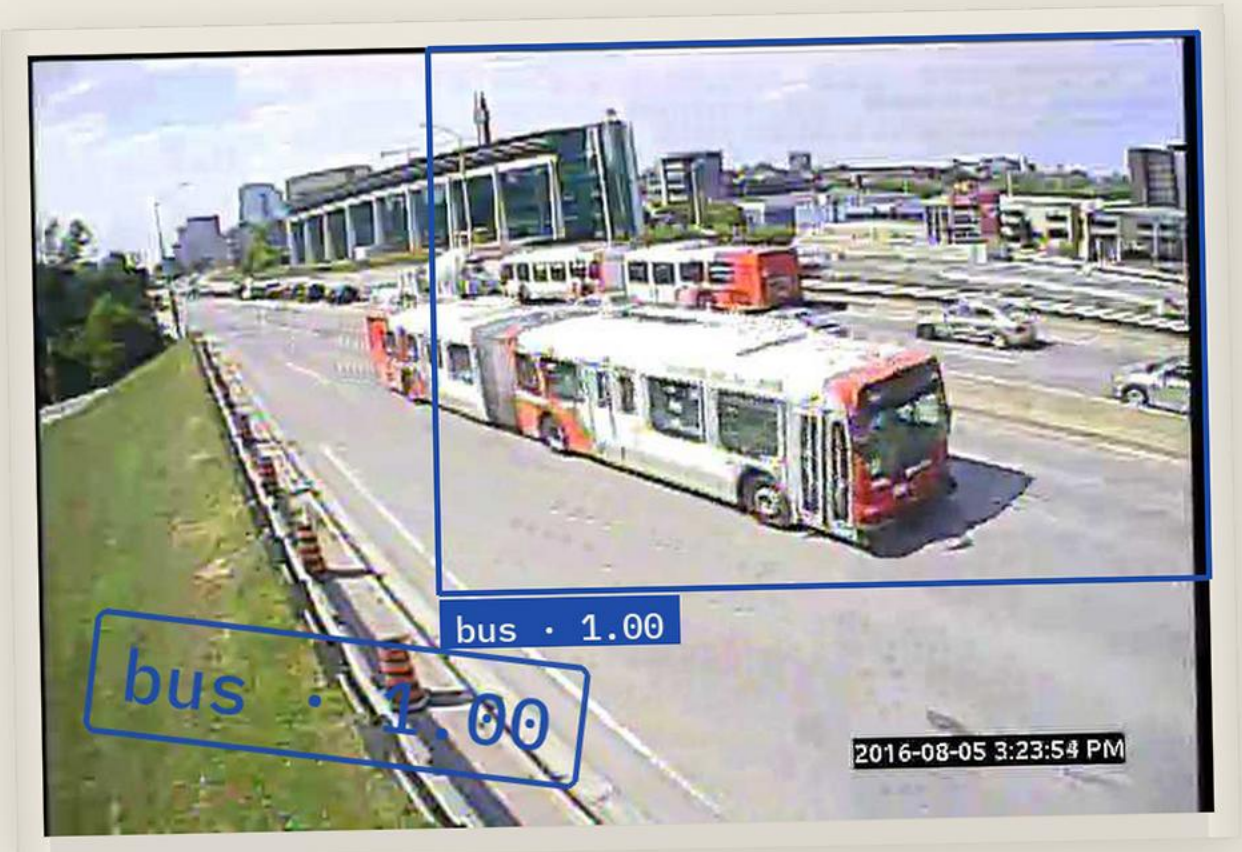
One frame, one box, one name.

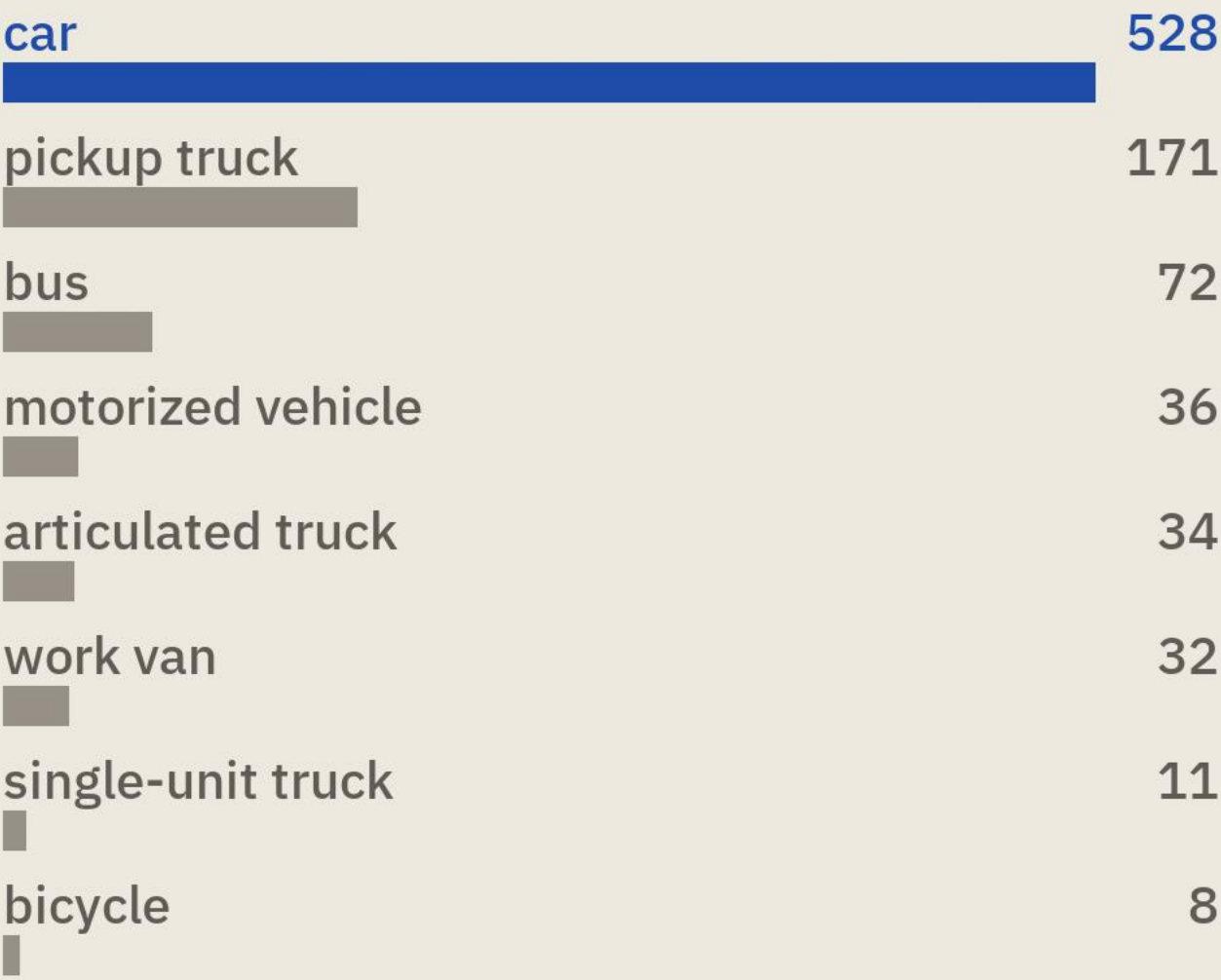


Maximus Friedrich Baluyot

Car is 59% of every frame.

Will it still see a bus?

900 validation frames by class · top 8 of 11 classes



Maximus Friedrich Baluyot

Vehicle detection

Capstone 1 · Intelligent transport systems

A box and a name in about 60 ms

```
max@baluyot ~ $ curl -s -F
file=@data/capstone1/Images/
00000001.jpg localhost:8101/predict
```



1 Frames and boxes
pandas · pillow

2 MobileNetV2 features
tensorflow

3 Class head + box head
tensorflow

4 POST /predict
FastAPI

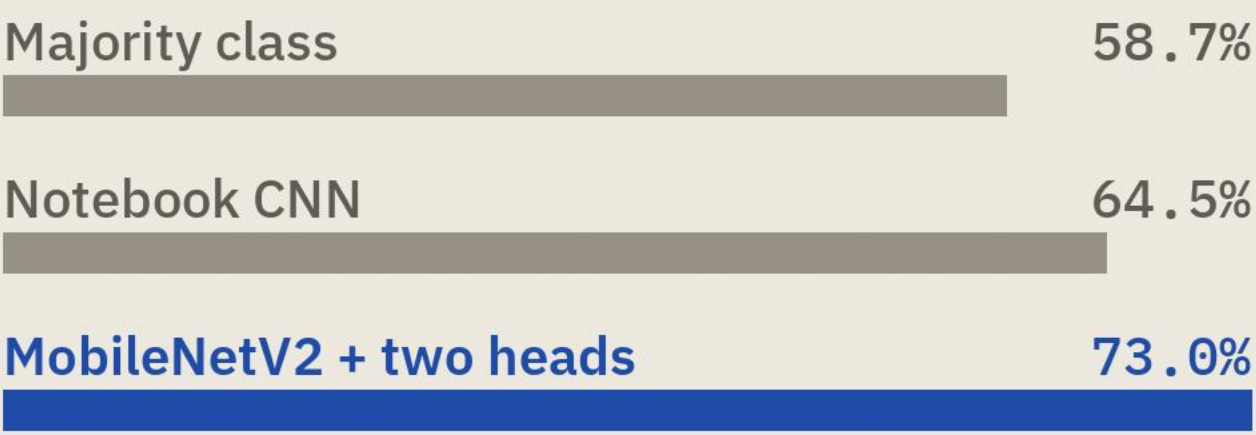
5 Two pages
HTML + ui.css

Weight rare classes so it won't just say car

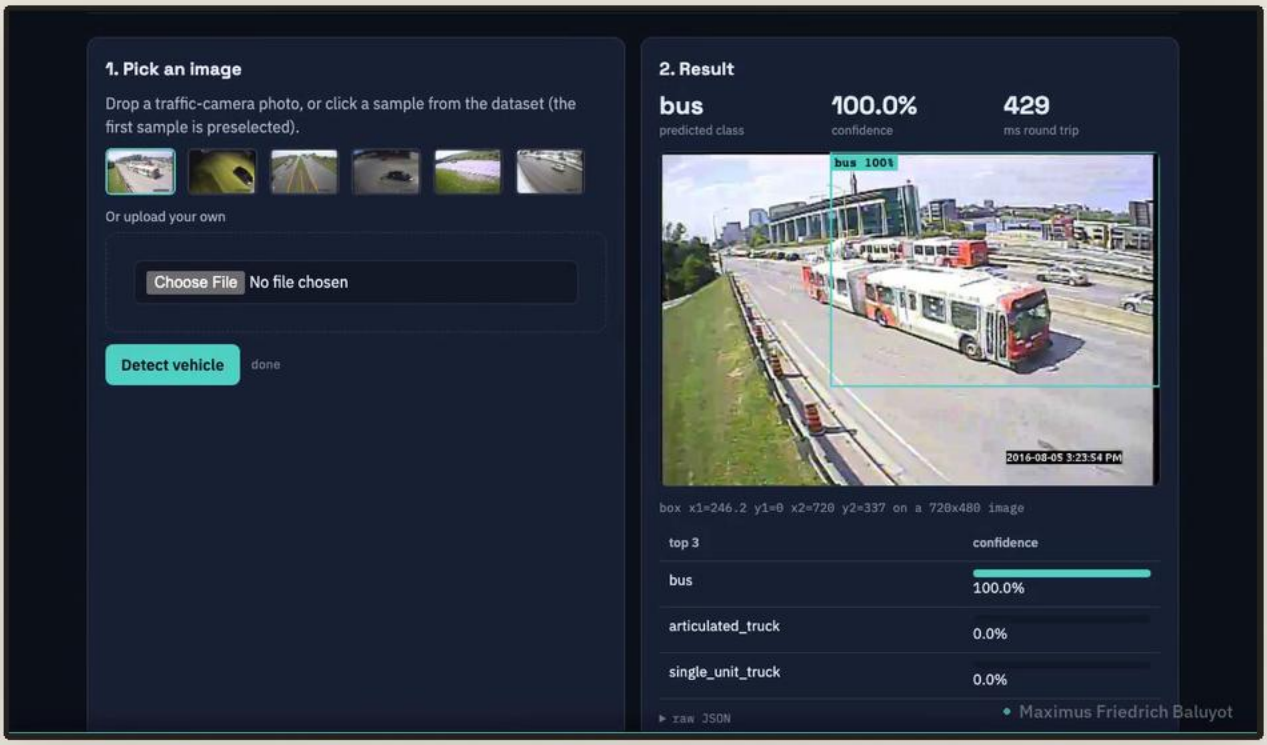
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3 models on the same holdout

val accuracy, 900 held-out frames



Detect: bus, one hundred percent, boxed



73.0%

val accuracy · 900 held-out frames · baseline 58.7%

one limit

One box per frame. IoU 0.25: boxes are loose.

Notebook 64.5 → 73.0 here. And it ships.

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Try it: sample 42, the near-empty road.

```
max@baluyot ~ $ git clone
https://github.com/Keshigami/
capstone1-vehicle-detection.git
max@baluyot ~ $ just setup
max@baluyot ~ $ just up
max@baluyot ~ $
```

github.com/Keshigami/capstone1-vehicle-detec...

Maximus Friedrich Baluyot

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